PNAS RESEARCH ARTICLE | BIOLOGICAL SCIENCES

Multiscale recoverable information in endogenous A-to-I RNA editing

Author list pending[a,1]

This manuscript was compiled on September 10, 2026

Endogenous adenosine-to-inosine RNA editing depends on duplex formation, cellular context, and nucleotide choice, but the information contributed at each scale remains unresolved. We used controlled held-out prediction to assay information recoverable from measured variables across 884 human inverted-Alu substrates. Whole-duplex sequence and predicted structure carried substantial signal and transferred strongly among human endpoints but unevenly across heterogeneous nonhuman datasets. Genomic inter-arm geometry added reproducible information about substrate-level editing burden beyond graph summaries, yet this propensity did not reliably improve site-level prediction. Matched sites varied across tissues, but tested RBP, transcript, repeat, and bulk chromatin annotations added no reproducible residual site-level information. Pair-first weighting improved learning from correlated sites, while a target-centered local representation organized by predicted pairing contributed information complementary to whole-duplex pooling; a matched-capacity shift control attributed the gain to correct target centering rather than generic capacity. Once these features were represented, richer supervision and model expansion produced limited or adverse returns under the tested regime. A frozen predictor implementing the supported principles, ADARPro, improved AUROC, AUPRC, F1, and Brier score in all five held-out endpoints. These experiments map information recoverable at substrate and nucleotide scales while locating an unresolved gap between current bulk measurements and site-level outcomes.

Significance

A-to-I RNA editing helps cells tolerate endogenous double-stranded RNA, yet why one RNA duplex or adenosine is edited while another is not remains incompletely understood. We used controlled prediction to ask which measured information is recoverable at each biological scale. Whole-duplex sequence and predicted structure captured broad signal; genomic organization carried additional signal for substrate-level editing propensity; and target-centered local context organized by predicted pairing added complementary site-level information. Tested bulk tissue and cell-line annotations did not recover reproducible additional site-level signal. These results define a conditional information map for present datasets and motivate matched donor-, cell-, transcript-, and structure-resolved measurements.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA), diversifying transcripts while preventing endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Yet potential substrates are not uniformly edited. Nearby adenosines in one predicted helix can differ sharply, and the same locus can differ among tissues (11, 12). Local sequence preference, mismatches, bulges, and helix context are established contributors (13–17), but they do not by themselves specify whether a transcript forms and retains the relevant duplex, occupies the appropriate compartment, or encounters active ADAR.

This selectivity suggests an information hierarchy. Whole-duplex sequence and folding constrain intrinsic substrate suitability; transcript- and cell-specific state may govern whether that substrate forms and remains accessible; and target-local context distinguishes adenosines within an available duplex. These levels have not been compared within one controlled predictive framework. Consequently, it remains unclear which aspects of endogenous editing are recoverable from existing measurements, which are obscured by measurement scale, and where additional model complexity ceases to help.

Prediction cannot establish mechanism, but controlled held-out prediction can assay biological information recoverable from measured variables under a specified representation, sample size, and evaluation regime. If a candidate variable improves prediction beyond

Author affiliations: [a]Affiliations pending author confirmation

Author contributions pending author confirmation.

Competing-interest declaration pending author confirmation.

Corresponding author and e-mail pending author confirmation.

a fixed sequence–structure reference, it carries reproducible information absent from that reference. Failure bounds only the tested measurement and formulation; it may reflect redundancy, noise, finite data, or model misspecification. Additions, ablations, transfer tests, and saturation experiments can therefore delineate what present data capture without treating association as causality.

AdarEdit established that a compact graph representation of sequence and predicted RNA structure can predict site-level editing in a pair-partitioned benchmark (18). We used that benchmark and model as a controlled reference to ask a different question: which information scales account for its signal, what adds reproducible information beyond it, and where additional measurements or model complexity stop helping. Whole-duplex sequence and predicted structure captured broad intrinsic signal; genomic organization carried additional pair-level burden signal; and a target-centered local representation recovered information lost by global pooling. Bulk contextual measurements did not close the remaining gap. The resulting predictor is an operational validation of this information map, not its principal scientific conclusion.

Results

A multiscale framework separates substrate, cellular, and target information. Endogenous editing can be interrogated at three distinguishable scales: intrinsic suitability of the RNA duplex, transcript and cellular context, and selection of an adenosine within it (Fig. 1A). We therefore asked how much whole-duplex sequence and predicted structure recover, whether substrate organization adds information, whether current contextual measurements recover residual variation, which target-centered information is lost by global representations, and where additional information or model capacity ceases to help.

We analyzed 884 graph-buildable inverted-Alu substrates across four individual GTEx tissues and Combined, a pooled endpoint. Complete Alu pairs were assigned without overlap to 566 training, 139 validation, and 179 test pairs. Discovery and model selection used pair-disjoint out-of-fold comparisons with pair-clustered uncertainty; the complete predictor and analysis manifest were frozen before the canonical held-out evaluation (Fig. 1B and C). Label definitions, fold isolation, evidence-status criteria, and evaluation safeguards are detailed in Methods and SI Appendix.

Sequence and predicted structure recover broad but non-universal signal. Whole-duplex sequence and predicted structure recover substantial editing signal across tissues and lineages. In retained benchmark characterization runs, the compact sequence–structure baseline achieved within-context AUROC of 0.869–0.933 across the five human endpoints and 0.823–0.865 across three nonhuman datasets (Fig. 2A). These values are descriptive across datasets rather than estimates of a common effect, but they establish that intrinsic RNA information is informative beyond one species or tissue.

Transfer delineated the scope of that signal. Models transferred strongly among human endpoints, whereas transfer among nonhuman datasets was more variable and human–nonhuman transfer was lower and asymmetric (Fig. 2B and C). The results support sequence–structure regularities that transfer across contexts but not universally; the present datasets cannot separate contributions from lineage, substrate composition, assay, labeling, and cellular environment.

High discrimination does not imply biological completeness. A sequence–structure model can rank sites while leaving substrate availability, tissue state, and unmeasured conformations unresolved. Subsequent tests therefore measured information added beyond this baseline rather than treating accuracy as variance explained.

Substrate organization reveals an editing-propensity layer. Substrate organization carries information about editing burden that the RNA graph does not encode explicitly. Genomic separation between Alu arms—discarded when model inputs join them with a fixed separator—correlated negatively with edited-site fraction in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 3B). Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 3C; SI Appendix). This establishes a reproducible pair-level signal beyond the tested graph-derived summaries; its dominant variable, physical inter-arm distance, is absent from the site-level graph input. The result marks substrate-level editing propensity, not measured duplex realization or ADAR engagement.

Tissue variation is evident, but current bulk measurements do not recover it. Editing outcomes varied among tissues even when genomic site identity was fixed. In a post-hoc analysis of development pairs, 1,221 of 74,758 sites observed in at least two individual tissues switched strictly between edited ($\geq 15\%$) and nonedited ($< 1\%$) states; among 50,162 sites observed in all four tissues, 17.4% spanned at least 15 EditingIndex points. Yet Alu-pair rankings on identical site sets remained correlated across every tissue pair ($\rho = 0.727$–0.921 for mean EditingIndex; SI Appendix, Fig. S7). Stable substrate differences therefore coexist with tissue-dependent site outcomes.

Predicted substrate burden did not translate reliably into better site-level classification. A cross-fitted gate added 0.0084 AUPRC but only 0.0041 F1, whose interval included zero, and increased Brier score by 0.0051; a neural burden auxiliary likewise failed its prespecified criterion. Static geometry therefore captures pair-level propensity without identifying why individual sites depart from it.

Biologically motivated contextual measurements likewise did not add reproducible residual site-level information. We tested replicated ENCODE eCLIP occupancy, sequence-based RBP predictions, tissue-average abundance and their interactions, with randomized or shuffled controls (19–22). Sequence-based predictions complemented eCLIP, for which Alu multimapping makes peak absence ambiguous. Neither cell-line occupancy nor tissue-average abundance improved the reference reproducibly. Cross-tissue label switches were nearly separated by tissue identity alone, so four tissue means could not identify an RBP-specific transferable effect. These results reject the tested proxies, not established RBP regulation (23–26).

Transcript and repeat annotations likewise failed to improve prediction beyond the graph. Exact-tissue

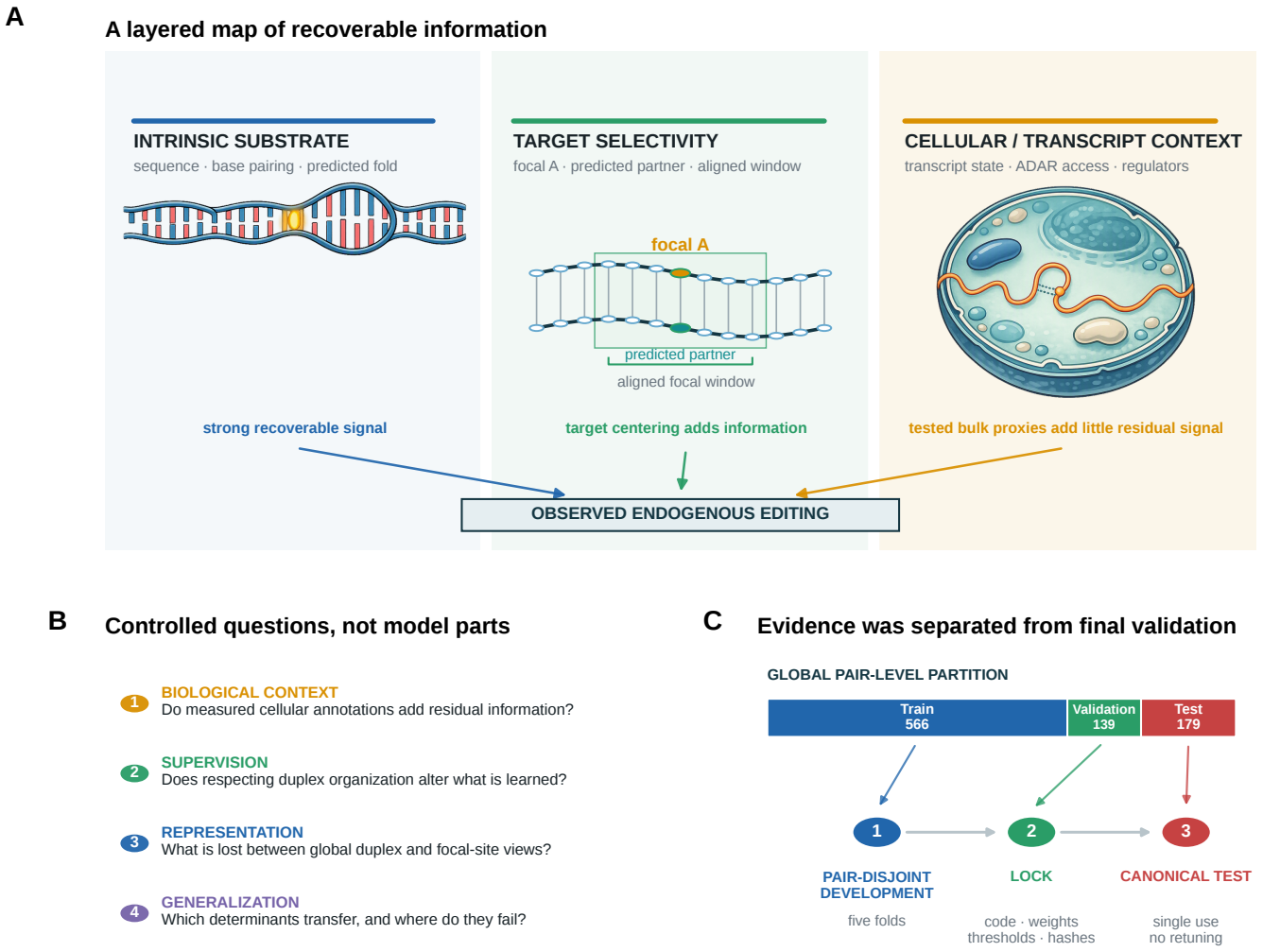


Fig. 1. An information hierarchy and leakage-safe strategy for mapping endogenous editing. (A) Intrinsic substrate properties, target-centered local context organized by predicted pairing, and transcript/cellular context are candidate information scales for observed editing; pair burden is an indirect readout of substrate propensity. Illustrations are conceptual. (B) Controlled comparisons interrogated context, supervision, representation, and generalization. (C) Complete pairs were partitioned globally, development used pair-disjoint folds, and the final specification was frozen before held-out evaluation. Combined is pooled and received no tissue-specific state vector.

bulk H3K36me3 overlap produced a small liver AUPRC increment without stable AUROC, F1, or pair-level benefit (27, 28). Thus substrate-level propensity is detectable, but the tested static, cell-line, and tissue-average measurements do not recover the residual site-level variation. Whether that variation reflects cellular state, measurement noise, finite data, or representation error is not determined here.

Pair-first weighting prevents site-rich substrates from dominating learning. Sampling substrates before sites improves prediction independently of adding labels. The canonical row set contains only part of the available high- and low-editing observations, and uniform site sampling gives each substrate influence proportional to its number of observations. Sites within one duplex share sequence, structure, genomic context, and often editing behavior, so that weighting is poorly matched to their dependence.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 4A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that expanded pool fixed and drawing a pair before class and site contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). Thus label recovery and substrate weighting make separable contributions.

The pair-first formulation improved the same-fold reference in all five endpoints (Fig. 4B); all 15 AUROC, AUPRC, and F1 intervals excluded zero. Within this benchmark, sampling pairs uniformly prevents site-rich substrates from receiving disproportionate influence during learning.

A descriptive pair-size audit localized this redistribution: pair-first learning improved log loss most for pairs with the fewest eligible sites, whereas the two largest site-count quintiles favored row-wise learning (SI Appendix, Fig. S8B).

Target-centered local information improves site-level prediction. A target-centered local representation organized by predicted pairing recovers site-level information diluted by whole-duplex pooling. Simple target-contact geometry

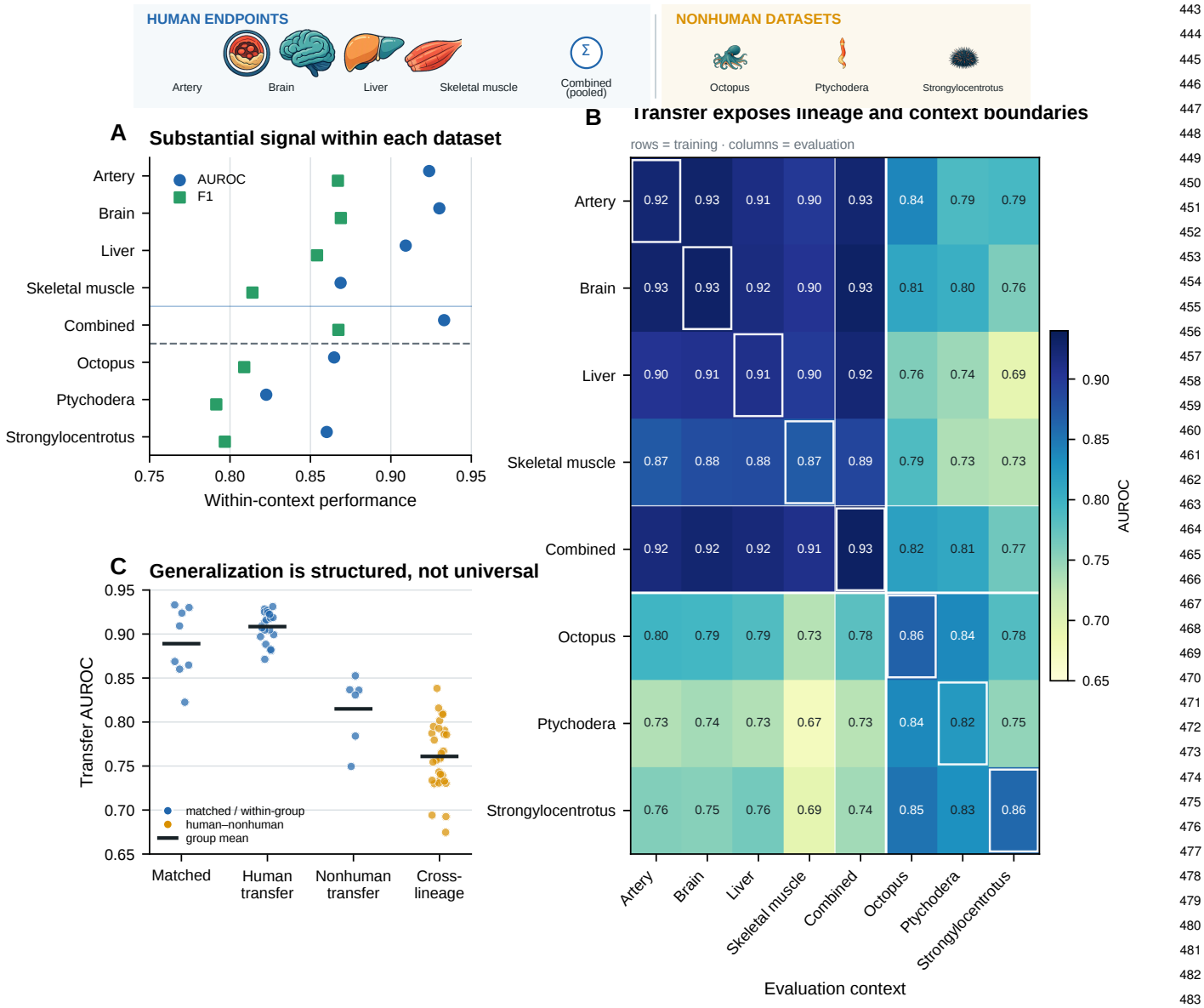


Fig. 2. Sequence–structure information is substantial but not universal. The tissue and lineage illustrations identify dataset classes and encode no measured values. (A) Within-context AUROC and F1 for the retained compact sequence–structure baseline across five human endpoints and three nonhuman datasets. Combined is pooled. (B) AUROC when each training context is evaluated in every context; white lines separate human and nonhuman blocks, and outlined diagonal cells mark matched-context evaluations. (C) Transfer values grouped by relation between training and evaluation contexts; horizontal bars are group means. These heterogeneous datasets define predictive scope but do not support causal attribution or meta-analysis.

improved ranking and calibration but not F1, and explicitly retaining the focal graph node improved AUROC and AUPRC more consistently than an otherwise matched duplicate-global control. Both results localized predictive information near the target before a specialized local encoder was built.

The selected local-augmented graph predictor adds a 65-position representation centered jointly on the focal adenosine and its dot-bracket-aligned helix partner. A low-capacity dilated convolution encodes nucleotide identity, pairing state, focal and partner indicators, and signed relative position; its output is fused with the whole-duplex graph embedding before classification. In muscle OOF data it reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 4C).

The decisive control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. This matched-capacity shift localizes the gain to the correctly centered, pairing-organized neighborhood rather than added convolutional capacity; it does not isolate the predicted partner from the rest of that neighborhood. Liver replication and the secondary natural-prevalence analysis showed the same pattern.

Descriptive structural stratification showed favorable Brier differences for paired and unpaired targets and across

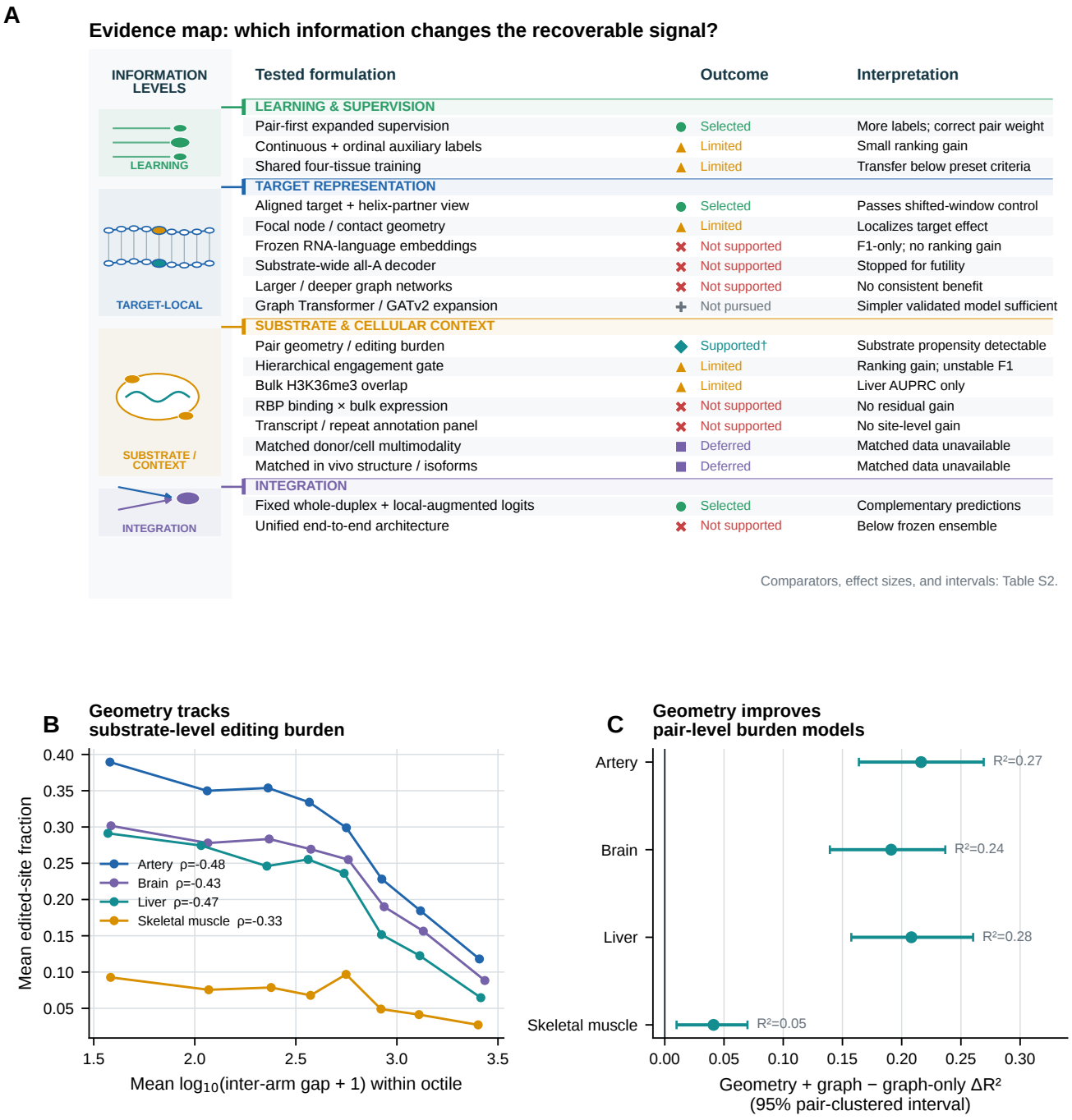


Fig. 3. A substrate-level signal remains unresolved by current context measurements. (A) Tested information sources grouped by scale and outcome; the left hierarchy maps directly to the evidence sections. SI Appendix, Table S2 gives each comparator, effect size, and interval. "Supported†" denotes a reproducible substrate-level observation not yet useful for site prediction; "not supported" applies only to the tested formulation, and deferred measurements were unavailable. (B) Mean edited-site fraction across octiles of inter-arm gap in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 from adding genomic geometry to graph summaries. Bars are 95% pair-clustered bootstrap intervals; labels give absolute R^2 for the combined model.

every target-stem extent, arguing against a trivial paired-status or stem-length explanation (SI Appendix, Fig. S8C and D).

Whole-duplex and target-local information remain complementary. The local-augmented predictor did not make an independently optimized whole-duplex view redundant. Their errors remained sufficiently distinct that a fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver; all six pair-clustered intervals excluded zero (Fig. 4D).

The final predictive specification consequently used two independently trained pair-first components: a whole-duplex-only graph predictor and a second graph predictor internally augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed 0.5/0.5 mean; the weights were not searched (Fig. 4E). Fixed integration preserved complementary predictive

views without fitting a flexible stacker to only hundreds of independent substrates.

In a post-hoc equal-pair analysis restricted to duplexes containing both labels, the fixed ensemble improved mean within-duplex AUROC over the pair-first component in muscle (+1.78 points; 95% pair-bootstrap interval +1.21 to +2.40) and liver (+1.38 points; +1.06 to +1.70). The aligned-local component alone showed a clear within-duplex increment only in muscle, so this analysis supports complementary within-substrate ranking rather than a replicated local-component effect (SI Appendix, Fig. S8A).

Richer supervision and model complexity yield diminishing returns. Once substrate-aware sampling and target-centered local information were represented, graded supervision and cross-tissue sharing produced only small ranking gains and adverse threshold or calibration tradeoffs. Both fell below their predeclared selection criteria (Fig. 3A; SI Appendix).

Broader representation and capacity changes likewise failed to improve the leading predictor reproducibly: substrate-wide decoding, frozen RNA-language embeddings, larger graph networks, and a preregistered connected integration were neutral or adverse on their prespecified comparisons (29) (SI Appendix). These results establish conditional saturation under the tested measurements, representations, and effective sample size; they do not define a universal prediction ceiling.

A frozen predictor generalizes to unseen benchmark pairs. After development selection, the complete predictor—components, checkpoints, thresholds, weights, code, and hashes—was frozen before canonical held-out evaluation.

The resulting predictor, ADARPro, improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 5A), with mean gains of 0.0277, 0.0261, and 0.0253 over reproduced supplied-checkpoint AdarEdit. Every endpoint had a positive 95% pair-clustered interval for all three metrics, and Brier score decreased throughout (Fig. 5B and C; SI Appendix, Table S1). Combined overlaps the individual tissue endpoints, so the mean is descriptive rather than a five-tissue meta-analysis. This result confirms that the frozen combination generalizes to unseen pairs within the benchmark; attribution to its individual components rests on the controlled development comparisons above.

The gains were not purchased by indiscriminate suppression of positive calls. At frozen validation-selected thresholds, ADARPro corrected 32.0–50.6% of baseline false positives while damaging 1.7–5.9% of baseline true positives (Fig. 5D). A secondary matched-recall analysis is reported in SI Appendix, Table S9.

Discussion

Endogenous A-to-I editing contains distinct recoverable information at the substrate and nucleotide scales. Whole-duplex sequence and predicted structure carry broadly transferable signal about intrinsic suitability; genomic organization marks a reproducible substrate-level editing propensity; and target-centered local context organized by predicted pairing contributes information obscured by whole-duplex pooling. Pair-first learning adds a statistical principle: correlated sites from one duplex should not acquire influence simply because that substrate contributes more rows.

This map connects observations that prior studies often treated separately. Sequence preferences and helix context are established determinants of local editability, whereas the distance between oppositely oriented Alus predicts aggregate Alu editing (13, 16, 17). Perturbational studies further establish regulation by RBPs and chromatin-associated processes (23–27). Our results place these findings on different information scales: intrinsic RNA and target context are recoverable from the benchmark, but cell-line occupancy and tissue-average abundance or chromatin do not identify regulator exposure or substrate state on an individual transcript. This failure constrains the tested measurement formulations; it does not imply absence of the underlying biology.

The remaining boundary is empirical and conditional. Once whole-duplex and target-centered information were represented and substrate dependence was respected, added supervision, generic RNA representations, model capacity, and joint integration produced diminishing or adverse returns under the tested regime. Residual error may reflect donor- or cell-specific ADAR state, transcript processing, regulator occupancy, in vivo conformation, measurement error, finite data, model misspecification, or biological stochasticity; these experiments cannot apportion those sources. They therefore locate a gap between current bulk measurements and site-level outcomes without identifying that gap as one latent cellular state.

Several limitations define the scope of this conclusion. The primary benchmark comprises graph-buildable human inverted-Alu substrates and tissue summaries, not all editing substrates or donor-resolved measurements. The heterogeneous nonhuman datasets reveal transfer loss without separating lineage, substrate class, assay, or labeling effects. Predicted secondary structure is only a surrogate for conformations available to ADAR in vivo. The canonical task is balanced and excludes intermediate observations; natural-prevalence analyses are secondary, and eCLIP coverage is cell-line-specific and repeat-censored. The frozen test establishes generalization to unseen pairs from the same benchmark construction, whereas component attribution comes from adaptively selected, nonmultiplicity-adjusted development comparisons. The upstream study reports cross-cohort label concordance, but the present predictor was not evaluated on an independently generated human cohort, so transportability remains untested (18).

The frozen predictor demonstrates the practical value of the information map by improving held-out discrimination, classification, and calibration. The broader result is a measurement agenda: future tests should pair editing with transcript abundance and processing, splicing and intron retention, ADAR isoform and interferon state, enzyme activity and localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation in the same donors or cells and on the same substrates. Such data could test, rather than assume, whether cellular state governs realization of a suitable duplex while local RNA context governs selection of particular adenosines.

Materials and Methods

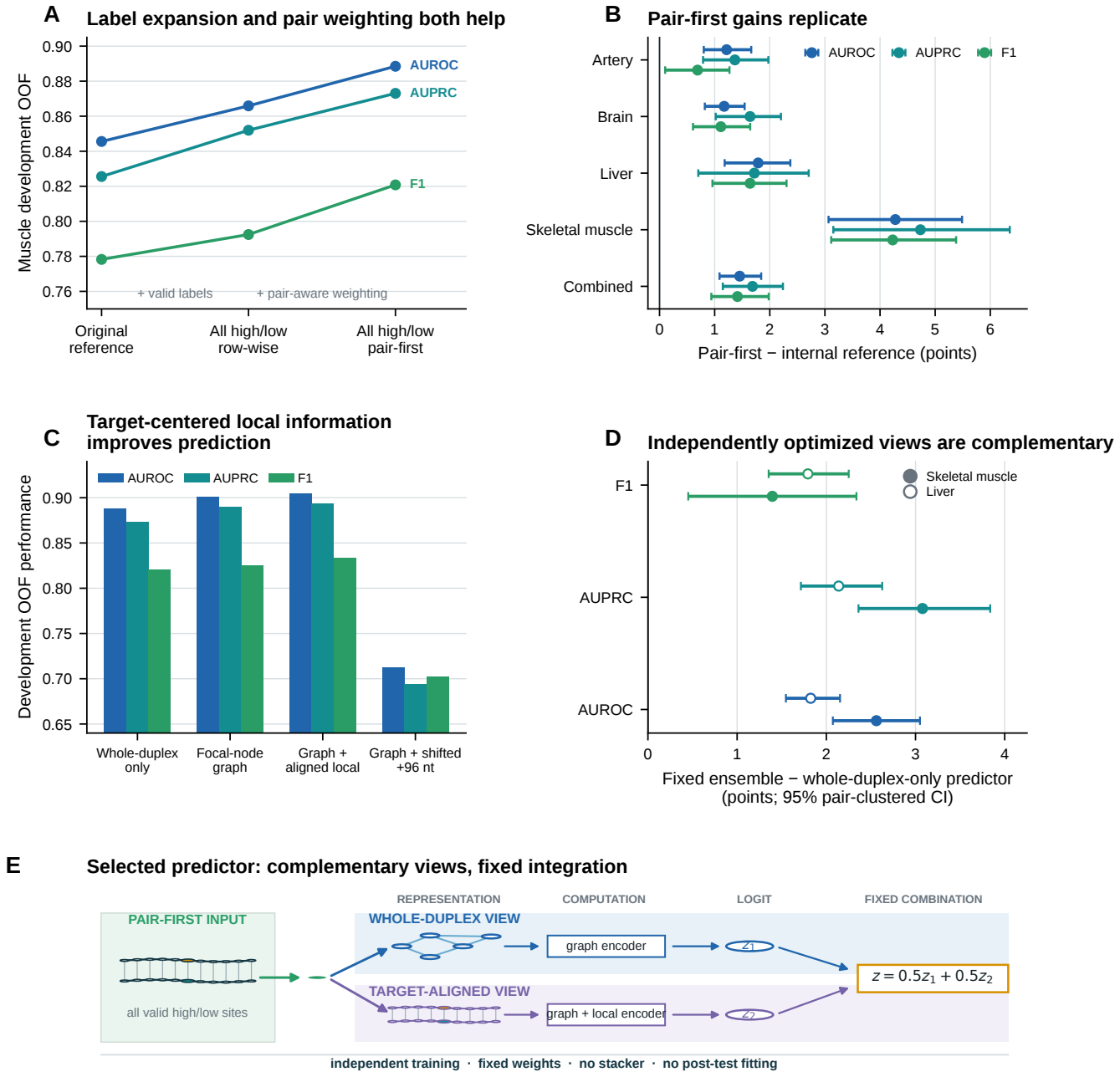


Fig. 4. Pair-first learning and target-centered local representation provide complementary predictive gains. (A) Same-fold ablations separate recovery of additional valid high/low labels from pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, focal-state retention, and aligned-local fusion form an evidence ladder; an otherwise identical local model shifted by 96 nt fails. (D) A fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) The operational predictor combines independently trained whole-duplex-only and graph-plus-local models without searched weights or post-test fitting. The RNA illustration is conceptual. All panels show development evidence.

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human labels were edited for EditingIndex $\geq 15\%$, nonedited for EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. Pair identities and the fixed 566/139/179 train/validation/test assignment were verified before analysis. Combined was treated only as its published pooled endpoint.

Development protocol. All discovery used the original train plus validation pairs divided into 5 outer folds by pair ID. Learned preprocessing, feature selection, checkpoint choice, and threshold choice were fitted without the outer fold. Predictions were concatenated only after every row was scored by a model that had not trained on its pair. The canonical test was inaccessible during development. Development OOF metrics estimate same-fold incremental effects; supplied-checkpoint test metrics remain the authoritative reference.

Models. The first component retained the compact AdarEdit graph attention encoder and trained from the expanded high/low site pool using uniform pair-first and equal-class sampling. The second, independently trained component retained the same graph encoder and added a 65-position target/helix-partner-aligned representation encoded by two dilated convolutional layers; local and graph embeddings were fused before its prediction head. The selected probability was $\sigma[(\operatorname{logit} p_{\text{whole}} + \operatorname{logit} p_{\text{augmented}})/2]$; weights were fixed at 0.5/0.5 rather than searched. The shifted-window control preserved the graph-plus-convolution architecture and optimization

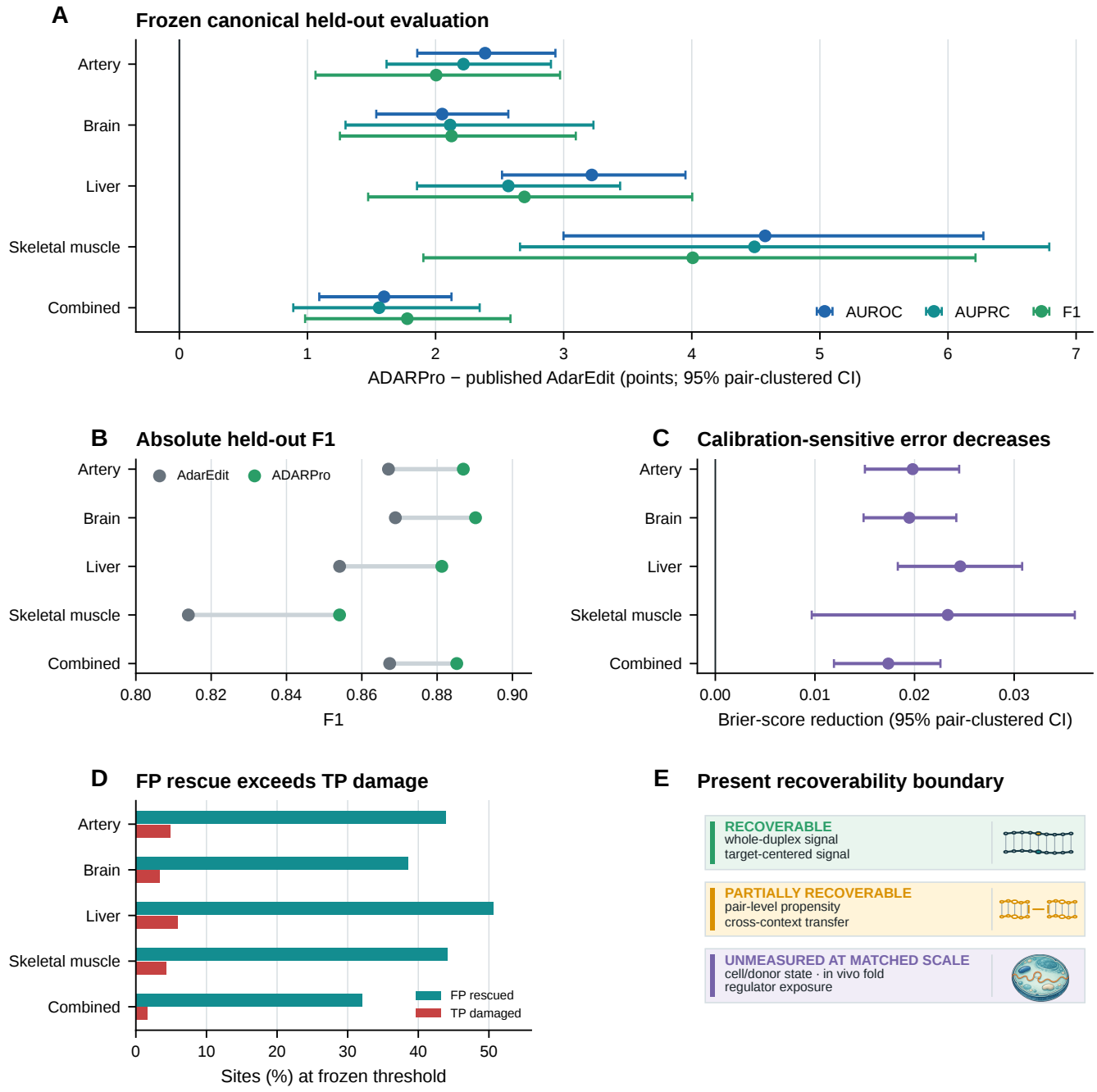


Fig. 5. Frozen evaluation confirms benchmark generalization and summarizes the present recoverability boundary. (A) ADARPro minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score. (D) Baseline false positives corrected and baseline true positives damaged at frozen validation-selected thresholds. (E) Information captured, partially resolved, and unavailable at matched biological scale; tier-specific icons are conceptual. Combined is pooled. All quantitative panels use the frozen canonical held-out evaluation.

while moving the local center 96 nt. Full feature definitions, architectures, ablations, and biological-context protocols are in SI Appendix.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each frozen training run. Uncertainty used a pair-clustered bootstrap with 2,000 replicates that resampled Alu pairs and retained all their sites. Development intervals quantify paired effect uncertainty and were not treated as multiplicity-adjusted confirmatory tests; selection also required recorded decision criteria and controls. Code, inputs, weights, epochs, thresholds, and checkpoint SHA-256 values were frozen before one held-out evaluation, with no subsequent model selection or tuning. Natural-prevalence and matched-recall analyses are secondary and described in SI Appendix.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. GPT Image 1.5 generated only label-free, nonquantitative figure components; prompts, candidate selection, and hashes are disclosed in the repository. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

1. B. L. Bass, RNA editing by adenosine deaminases that act on RNA. *Annu. Rev. Biochem.* **71**, 817–846 (2002).
2. K. Nishikura, A-to-I editing of coding and non-coding RNAs by ADARs. *Nat. Rev. Mol. Cell Biol.* **17**, 83–96 (2016).
3. E. Eisenberg, E. Y. Levanon, A-to-I RNA editing—immune protector and transcriptome diversifier. *Nat. Rev. Genet.* **19**, 473–490 (2018).
4. B. J. Liddicoat, et al., RNA editing by ADAR1 prevents MDA5 sensing of endogenous dsRNA as nonself. *Science* **349**, 1115–1120 (2015).
5. N. M. Mannion, et al., The RNA-editing enzyme ADAR1 controls innate immune responses to RNA. *Cell Reports* **9**, 1482–1494 (2014).
6. E. Y. Levanon, et al., Systematic identification of abundant A-to-I editing sites in the human transcriptome. *Nat. Biotechnol.* **22**, 1001–1005 (2004).
7. A. Athanasiadis, A. Rich, S. Maas, Widespread A-to-I RNA editing of Alu-containing mRNAs in the human transcriptome. *PLoS Biol.* **2**, e391 (2004).
8. D. D. Y. Kim, et al., Widespread RNA editing of embedded Alu elements in the human transcriptome. *Genome Res.* **14**, 1719–1725 (2004).
9. L. Bazak, et al., A-to-I RNA editing occurs at over a hundred million genomic sites, located in a majority of human genes. *Genome Res.* **24**, 365–376 (2014).
10. H. T. Porath, S. Carmi, E. Y. Levanon, A genome-wide map of hyper-edited RNA reveals numerous new sites. *Nat. Commun.* **5**, 4726 (2014).
11. M. H. Tan, et al., Dynamic landscape and regulation of RNA editing in mammals. *Nature* **550**, 249–254 (2017).
12. E. Picardi, et al., Profiling RNA editing in human tissues: towards the inosinome atlas. *Sci. Reports* **5**, 14941 (2015).
13. J. M. Eggington, T. Greene, B. L. Bass, Predicting sites of ADAR editing in double-stranded RNA. *Nat. Commun.* **2**, 319 (2011).
14. S. K. Wong, S. Sato, D. W. Lazinski, Substrate recognition by ADAR1 and ADAR2. *RNA* **7**, 846–858 (2001).
15. K. A. Lehmann, B. L. Bass, Double-stranded RNA adenosine deaminases ADAR1 and ADAR2 have overlapping specificities. *Biochemistry* **39**, 12875–12884 (2000).
16. L. Bazak, E. Y. Levanon, E. Eisenberg, Genome-wide analysis of Alu editability. *Nucleic Acids Res.* **42**, 6876–6884 (2014).
17. M. S. Zambrano-Mila, et al., Dissecting the basis for differential substrate specificity of ADAR1 and ADAR2. *Nat. Commun.* **14**, 8212 (2023).
18. Z. Rosenwasser, M. Levitt, E. Y. Levanon, G. Oren, Structure-aware graph learning predicts RNA editability across tissues and species (bioRxiv preprint) (2026).
19. E. L. Van Nostrand, et al., A large-scale binding and functional map of human RNA-binding proteins. *Nature* **583**, 711–719 (2020).
20. L. Sun, et al., Predicting dynamic cellular protein–RNA interactions by deep learning using in vivo RNA structures. *Cell Res.* **31**, 495–516 (2021).
21. M. Horlacher, et al., Towards in silico CLIP-seq: predicting protein–RNA interaction via sequence-to-signal learning. *Genome Biol.* **24**, 180 (2023).
22. GTEx Consortium, The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science* **369**, 1318–1330 (2020).
23. E. C. Freund, et al., Unbiased identification of trans regulators of ADAR and A-to-I RNA editing. *Cell Reports* **31**, 107656 (2020).
24. G. Quinones-Valdez, et al., Regulation of RNA editing by RNA-binding proteins in human cells. *Commun. Biol.* **2**, 19 (2019).
25. E. J. Park, et al., Primate lineage specification requires suppression of Alu hyperediting (bioRxiv preprint) (2026).
26. Y. Yi, et al., A dual role of EZH2 in regulating A-to-I RNA editing and mRNA stability through ADAR. *Nat. Commun.* **17**, 4421 (2026).
27. P. Cai, et al., Chromatin state shapes site-specific A-to-I RNA editing (bioRxiv preprint) (2026).
28. ENCODE Project Consortium, Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature* **583**, 699–710 (2020).
29. R. J. Penic, T. Vlasic, R. G. Huber, Y. Wan, M. Sikic, RiNALMo: general-purpose RNA language models can generalize well on structure prediction tasks. *Nat. Commun.* **16**, 5671 (2025).